SUMMARY : INTRODUCTION TO MS EXCEL-III

SESSION OVERVIEW:

By the end of this session, the students will be able to:

- Perform different **statistical, logical operator, string, and date functions** in Ms Excel.
- Understand different **data AI tools** in MS Excel.

KEY TOPICS AND EXAMPLES:

1. Perform different functions in MS Excel:

Reference datasets: (dataset-1) and (dataset-2)

- **Mathematical Calculations**: Functions like SUM, AVERAGE, MAX, and MIN are used for basic arithmetic operations and statistical calculations.
- **Statistical Analysis:** Functions like COUNT, COUNTA, COUNTIF, and SUMIF are used for counting and summing data based on specified conditions, as well as other statistical calculations.
- **Logical Operations:** Functions like IF, AND, OR, and NOT are used to perform logical operations and make decisions based on specified criteria.
- **Text Manipulation:** Functions like CONCATENATE, LEFT, RIGHT, and MID are used to manipulate and concatenate text strings. These operations are majorly used in data cleaning before the data analysis.
- **Date and Time Operations**: Functions like TODAY, NOW, DATE, and TIME are used to work with date and time values and perform calculations based on them.

a. Statistical functions:

Note: *Use the Product Recall (additional dataset) to explore and understand the application of statistical functions through examples.*

i. **COUNT:** Counts the number of cells containing numbers in a range.

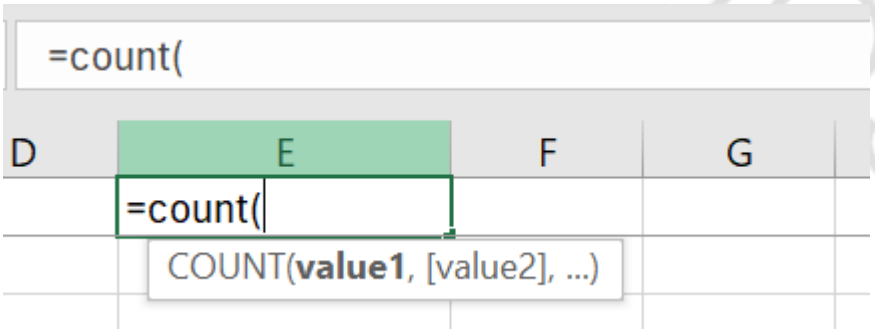


Figure 21: *Representation of COUNT in Excel*

Example :
You are working with a dataset called Product Recall, which contains information about different products that were recalled from the market. One of the columns in the dataset is Units Recalled, which lists how many units of each product were recalled. Your task is to find the total number of entries (rows) where the recall units are recorded.

Solution:

```
=COUNT(start_cell:end_cell)
=COUNT(F3:F3002)
```

(The output will give us the total number of numeric entries in the "Units Recalled" column within the range F3 to F3002).

ii. **COUNTA:** Counts the number of non-empty cells in a range.

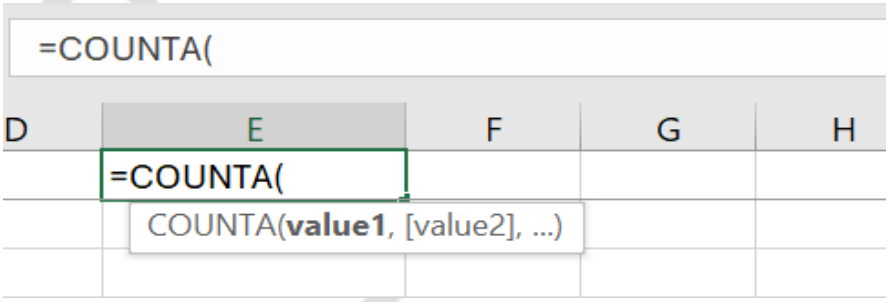


Figure 22: Representation of COUNTA in Excel

iii. **COUNTIF:** Counts the number of cells that meet a specific condition.

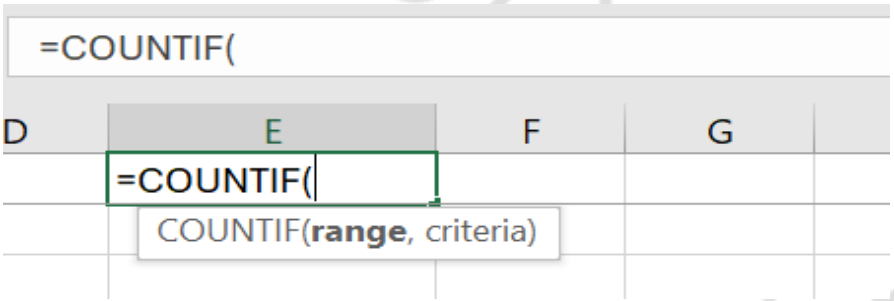


Figure 23: *Representation of COUNTIF in Excel*

Example :
You are analyzing the Product Recall dataset, which contains details such as product IDs, defect rates, recall reasons, and the number of units recalled.
Now, you want to find out how many products were recalled due to a "Defective Product".

Solution:

```
=COUNTIF(range, criteria)
=COUNTIF(G3:G3002, "Defective Product")
```

(COUNTIF goes row by row through the "Reason for Recall" column, checks which ones are labeled "Defective Product," and then gives us the total count of such rows.).

iv. **SUMIF:** Adds up the numbers in a range that meets a specific condition.

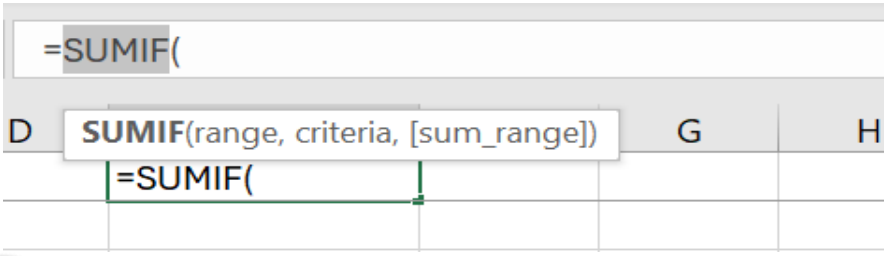


***Figure 24**: Representation of SUMIF in Excel*

Note:

Now we will use cell referencing, which makes formulas more dynamic and flexible:

- *If the value in the referenced cell changes, the result updates automatically.*
- *It reduces errors since you don't need to edit the formula each time.*
- *With $, you can lock a cell reference so it always points to the same place.*

Example :

You are analyzing the Product Recall dataset. Each row contains the Units Recalled and the corresponding Reason for Recall.

Your task is to find the total number of units recalled where the recall reason is Safety Issue.

Solution:

You can refer to any cell that contains Safety Issue

```
=SUMIF(G3:G3002,$G$11,F3:F3002)
```

How It Works:

- Excel checks the range G3:G3002 (Reason for Recall).
- For each row that matches the text in G5 (Safety Issue), it picks the corresponding value from F3:F3002 (Units Recalled).
- Finally, it adds up all those values to give the total units recalled due to safety issues.

b. **Logical operator functions: (dataset)**

Steps:

- Click on the cell where you want the result of the logical function to appear.

- Start typing the name of the logical function you want to use (e.g., IF, AND, OR).
- After typing the function name, open parentheses "(" to begin entering arguments.
- Enter the conditions or values you want to evaluate. Separate multiple arguments with commas.
- Close parentheses ")" to indicate the end of the function's arguments.
- Press Enter to complete the function. Excel will calculate the result based on the specified function and arguments.

i. **IF**: Returns one value if a condition is true and another value if it's false.

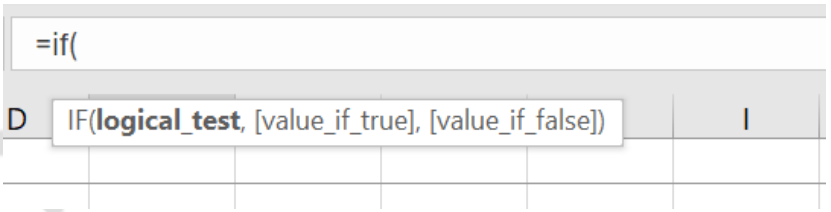


Figure 7: *Representation of IF function in Excel*

ii. **AND**: Returns TRUE if all arguments are true, otherwise returns FALSE.

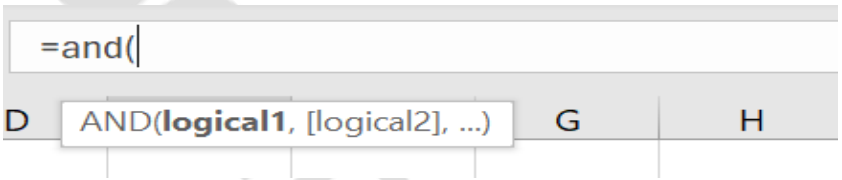


Figure 8: *Representation of AND function in Excel*

iii. **OR**: Returns TRUE if any argument is true, otherwise returns FALSE.

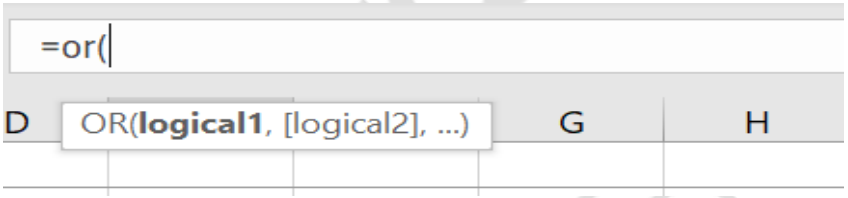


Figure 9: *Representation of OR function in Excel*

iv. **NOT**: Reverses the logical value of its argument (TRUE becomes FALSE, and FALSE becomes TRUE).

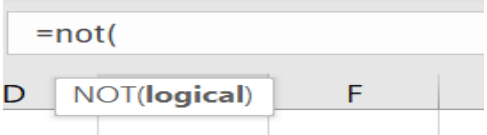


Figure 10: *Representation of NOT function in Excel*

Example :

You are checking whether a student is **not absent**.

Column B contains **TRUE** if the student is absent and **FALSE** if present.

Solution:
We use **NOT** :

```
=NOT(B2)
```

How It Works:

- NOT simply **reverses the value** - TRUE becomes FALSE, and FALSE becomes TRUE.

Example :

You are analyzing the Employee Performance dataset. Each employee has a Sales Score (column C) and Attendance % (column D).
Classify employees using the rules below:

- If Sales Score ≥ 80 **AND** Attendance ≥ 90% → label as **"Excellent"**
- Otherwise → label as **"Needs Improvement"**

Solution:
We use **AND** inside IF:

```
=IF(AND(C2>=80,D2>=90),"Excellent","Needs Improvement")
```

How It Works:

- The result is **"Excellent" only when both conditions are true**; otherwise, it returns **"Needs Improvement."**

Note:
Now we will use Nested IF, which helps handle multiple conditions step by step:

- *It allows you to check more than one condition in the same formula.*
- *It helps categorize data into multiple outcomes (e.g., Low, Moderate, Critical) instead of only two.*
- *By combining with AND/OR, you can create more complex decision rules.*

v. **Nested IF**: The IF function becomes even more powerful when you combine multiple IF statements together. This is called a nested IF. It allows you to check more than one condition and return different results accordingly.

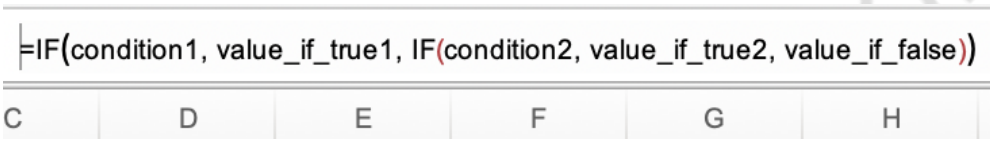


Figure 11: *Representation of Nested If function in Excel*

Example :
Suppose you want to categorize Defect Rate into “High”, “Medium”, and “Low.”

- If Defect Rate > 0.15 → “High”
- If Defect Rate > 0.08 → “Medium”
- Otherwise → “Low”

Formula (assuming Defect Rate is in cell E3):

Solution:

```
=IF(E3>0.15,"High",IF(E3>0.08,"Medium","Low"))
```

How it works:

- Here, the first IF checks if defect rate is greater than 0.15.
- If not, the second IF checks if it’s greater than 0.08.
- If neither is true, the result is “Low.”.

Example :
You are analyzing the Product Recall dataset. Each product has a Defect Rate (column E) and Units Recalled (column F).
Classify each product based on the following rules:

- If (Defect Rate > 0.15 AND Units Recalled > 500) OR (Defect Rate > 0.2 regardless of units) → label it “Critical”
- Else if Defect Rate > 0.08 → label it “Moderate”
- Otherwise → label it “Low”

Solution:
We can solve this with a nested IF using AND + OR

```
=IF(OR(AND(E3>0.15,F3>500),E3>0.2),"Critical",IF(E3>0.08,"Moderate","Low"))
```

How It Works:

- The first IF checks an OR condition:
  - AND(E3>0.15,F3>500) → Defect Rate above 0.15 and Units > 500
  - E3>0.2 → Defect Rate above 0.2 (regardless of units)
  - If either is true → result is “Critical.”
- If not, Excel moves to the second IF (E3>0.08).
  - If TRUE → returns “Moderate.”
- If neither condition is met → returns “Low.”

- **IFERROR :** The IFERROR function in Excel is a built-in function used to handle errors in calculations or lookups by returning a custom value when an error occurs. Instead of displaying standard error messages such as `#DIV/0!`, `#N/A`, or `#VALUE!`, the function allows users to replace them with a specified message or

default value.

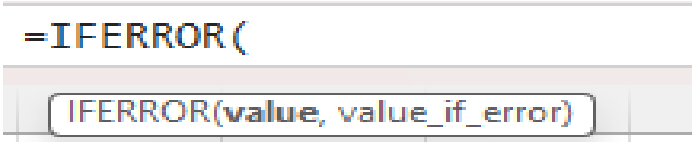


Figure 11: *Representation of IFERROR function in Excel*

Example: Using `IFERROR`, instead of displaying an error, it returns "Division not possible".

Formula *=IFERROR(10/0, "Division not possible")*

c. **Text functions: (Dataset)**

Steps:

- Click on the cell where you want the result of the text function to appear.
- Start typing the name of the text function you want to use (e.g., LEFT, RIGHT, CONCATENATE).
- After typing the function name, open parentheses "(" to begin entering arguments.
- Enter the text or cell reference you want to manipulate. Additional arguments may be required depending on the function.
- Close parentheses ")" to indicate the end of the function's arguments.
- Press Enter to complete the function. Excel will calculate the result based on the specified text function and arguments.

i. **CONCATENATE**: Combines multiple text strings into one.

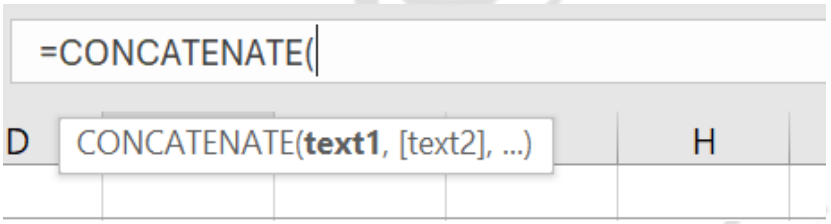


Figure 11: *Representation of CONCATENATE function in Excel*

ii. **LEFT:** Returns the leftmost characters from a text string.

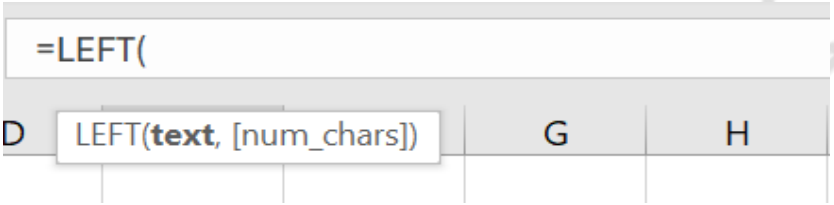


Figure 12: *Representation of LEFT function in Excel*

iii. **RIGHT:** Returns the rightmost characters from a text string.

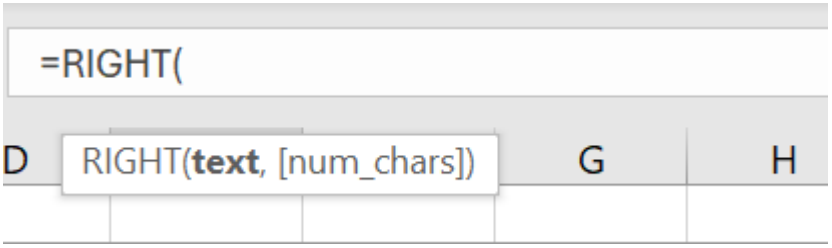


Figure 13: Representation of the RIGHT function in Excel

iv. **MID:** Returns a specific number of characters from a text string, starting at a specified position.

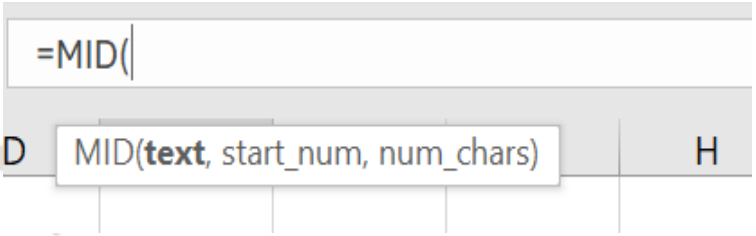


Figure 14: Representation of MID function in Excel

v. **TRIM:** Removes leading and trailing spaces from a text string.

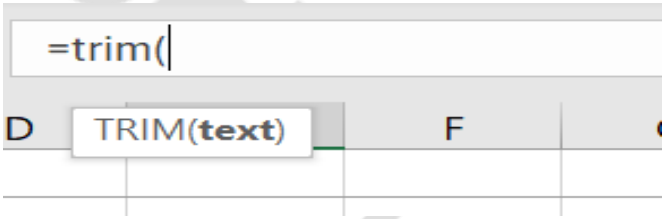


Figure 15: Representation of TRIM text function in Excel

vi. **LEN**: Returns the length of a text string (number of characters).

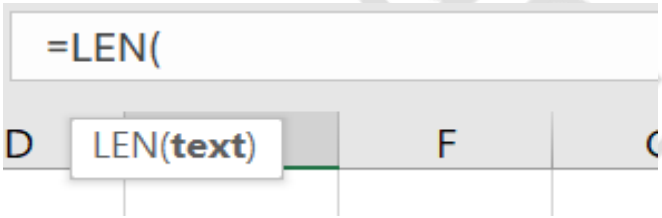


Figure 16: Representation of LENGTH of string function in Excel

vii. **TEXT:** Format numbers, dates, and other values into a specific text format.

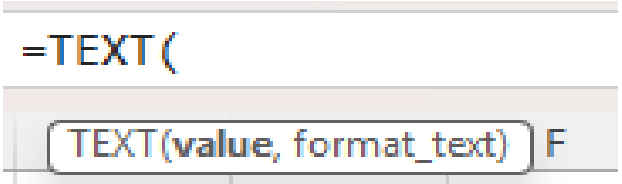


Figure 17: Representation of TEXT function in Excel

d. **Date functions: (Dataset)**

Note:

- *Excel stores dates as serial numbers, and date functions help convert or extract useful parts from them.*
- *Functions like YEAR, MONTH, and DAY let you pull specific components from a full date.*
- *Practice using these functions on different dates to clearly see how Excel interprets and calculates with them.*

i. **TODAY:** Returns the current date.

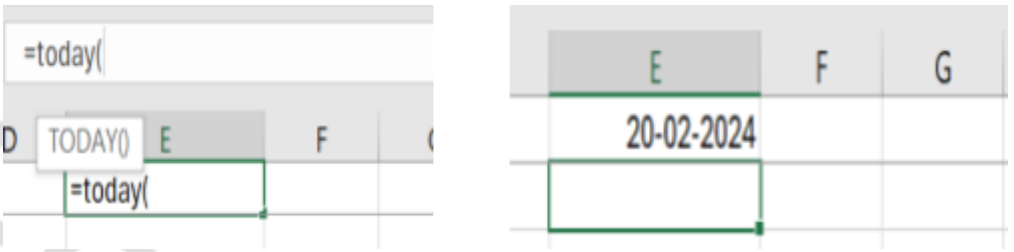


Figure 17: *Representation of TODAY function in Excel*

ii. **NOW:** Returns the current date and time.



Figure 18: *Representation of NOW in Excel*

iii. **DATE**: Creates a date based on the specified year, month, and day.



Figure 19: *Representation of DATE in Excel*

iv. **TIME:** Creates a time based on the specified hour, minute, and second.

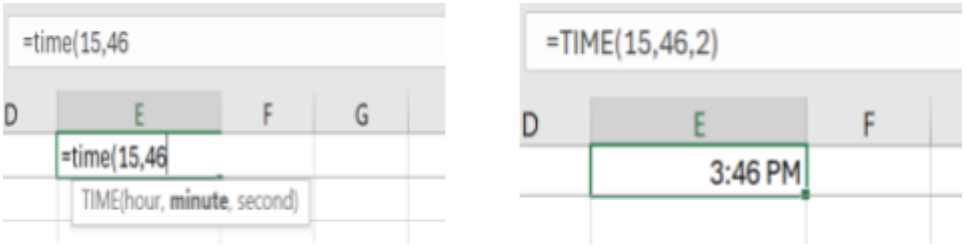


Figure 20: *Representation of TIME in Excel*

v. **YEAR:** Extracts the year from a date.

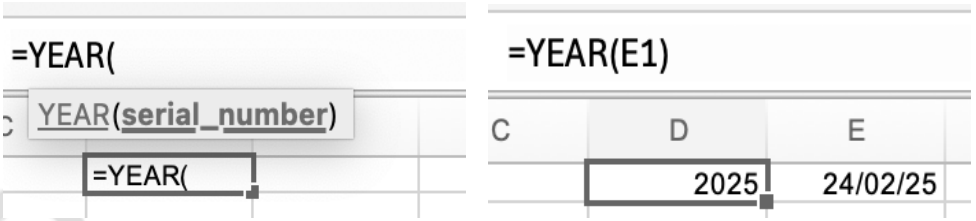


Figure 22: *Representation of YEAR in Excel*

vi. **MONTH:** Extracts the month number (1–12) from a date.

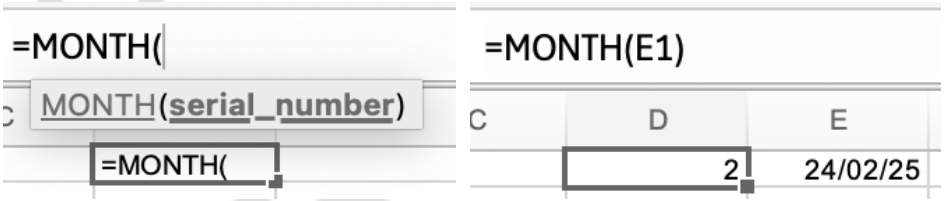


Figure 23: *Representation of MONTH in Excel*

vii. **DAY:** Extracts the day of the month (1–31) from a date.

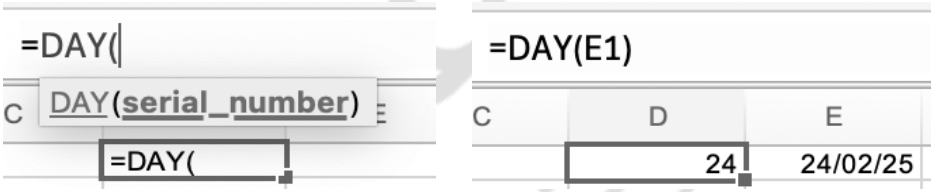


Figure 24: *Representation of DAY in Excel*

Note:

- *In Excel, date conversion changes only the display format, not the underlying stored value.*
- *Built-in and custom formats let you show the same date in multiple styles.*
- *The actual date value remains unchanged regardless of how it is displayed.*
- *Excel functions can also identify if a date falls on a weekday or weekend.*
- *This is useful for scheduling, planning, and data analysis.*

❖ **Date Format Conversion:**

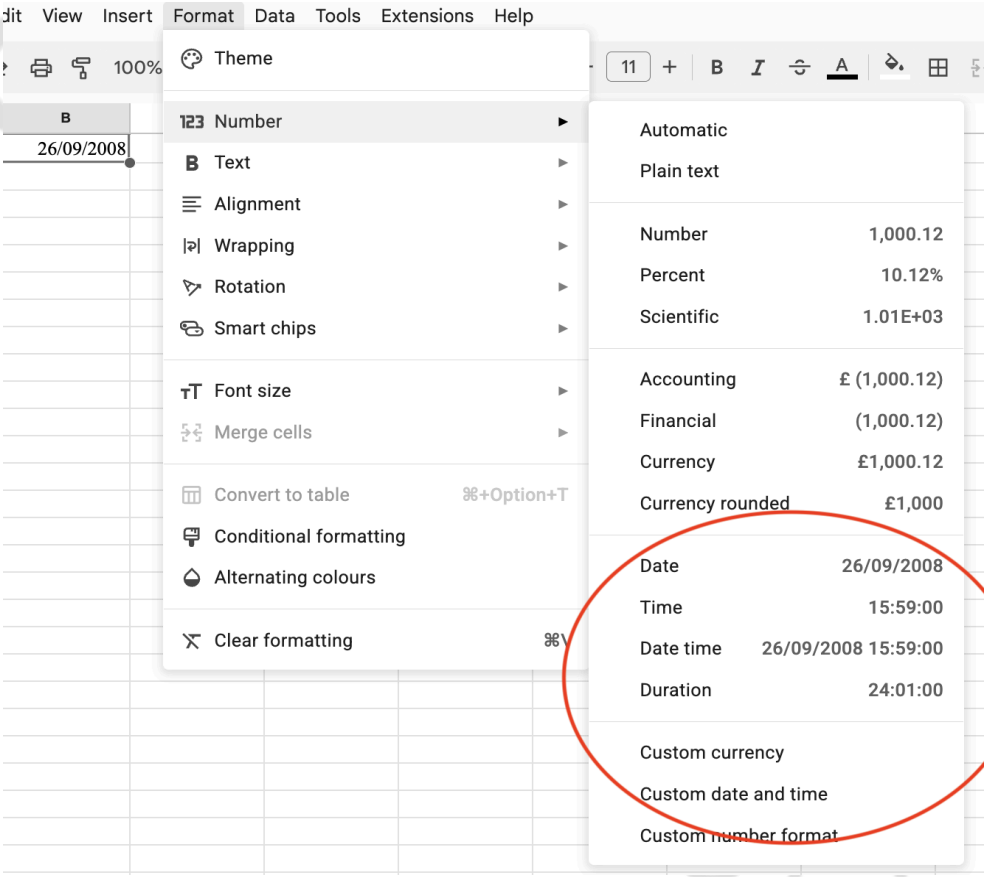
Excel stores dates as serial numbers, but you can change how they appear without affecting calculations.

Key Points:

- Format Cells: Right-click → Format Cells → Number → Date to choose common formats (e.g., DD-MM-YYYY, MMMM DD, YYYY).
- TEXT Function: Convert dates to custom text formats
    - Example: =TEXT(A2,"DD-MMM-YYYY") → 28-Aug-2025
- DATEVALUE: Converts a date stored as text into a valid Excel date.
    - Example: =DATEVALUE("15-Aug-2025") → 45144

Tip: Formatting only changes appearance, not the actual date value, which Excel uses for calculations.

1. **CUSTOM DATE FORMAT:** Converting date into different formats means applying a different display pattern while the underlying date remains the same.



How to do this:

1. Select the cell(s) with the date.
2. Right-click → Format Cells → Date.
3. Choose a desired format from the list (or create a custom one under "Custom").

Example:

Convert the date 13/02/2025 into different formats to see how Excel displays the same date in multiple ways.

Solution:

Different display styles of the same date (13/02/2025) using custom formats.

a. 13/02/25 → Format: dd/mm/yy (shows day, month, and 2-digit year)
b. 13 Feb 2025 → Format: dd mmm yyyy (shows month name in short form)
c. 2025-02-13 → Format: yyyy-mm-dd (year comes first, useful in databases/standard notation)
d. 13/02/2025 10:30 AM → Format: dd/mm/yyyy hh:mm AM/PM (displays both date and time)

2. **DATEVALUE:** Converts a date stored as text into Excel's internal serial number format. This number represents the total days counted since 1-Jan-1900, which Excel uses to perform date calculations.

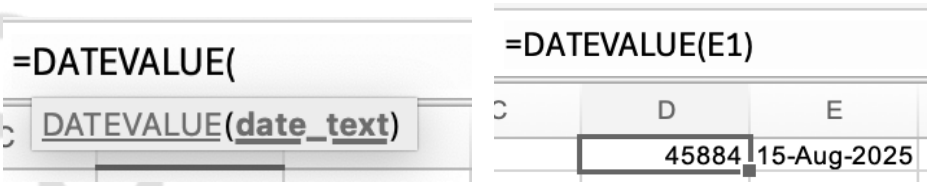


Figure 25: *Representation of DATEVALUE in Excel*

3. **TEXT:** Converts a date (or number) into text in a specified format, allowing you to display it exactly as you want (e.g., "DD-MMM-YYYY" or "MMMM YYYY"), without changing the underlying date value.

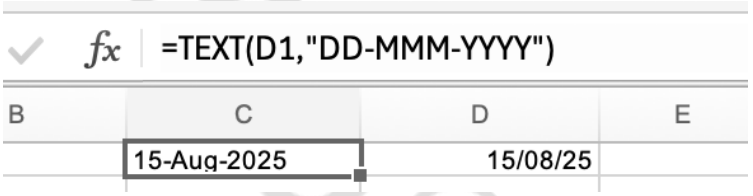


Figure 26: *Representation of Text Conversion in Excel*

4. **EXTRACTING WEEKDAY/WEEKEND INFORMATION:** Helps identify the day of the week from a given date (e.g., Monday, Tue) or check if a date falls on a weekday or weekend, useful for scheduling and analysis.

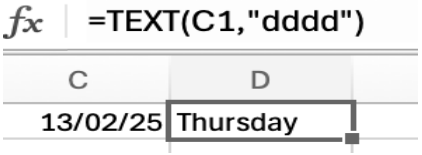


Figure 27: *Extracting day of date*

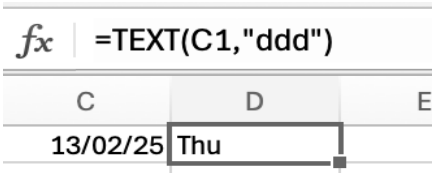


Figure 28: *Extracting short form of day from date*

➔ To check if a date falls on a weekday or weekend

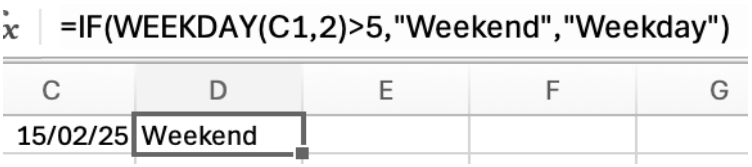


Figure 29: *Extracting Weekday or Weekend from date*

How it works:

- WEEKDAY(A1,2) → returns 1–7 for Mon–Sun (e.g., 15/02/2025 = 6).
- >5 → checks if it's 6 or 7 (i.e., Sat/Sun).
- IF(...) → TRUE ⇒ "Weekend", FALSE ⇒ "Weekday".

AI FEATURES IN EXCEL: (Dataset)

1. Analyze data:

This feature in Excel gives insights into the data based on the prompt. This is an AI feature. This feature can help us identify outliers in the data just by asking questions.

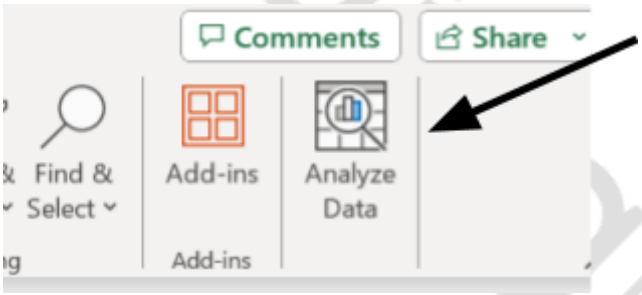


Figure 27*: Represents Analyze data AI feature*

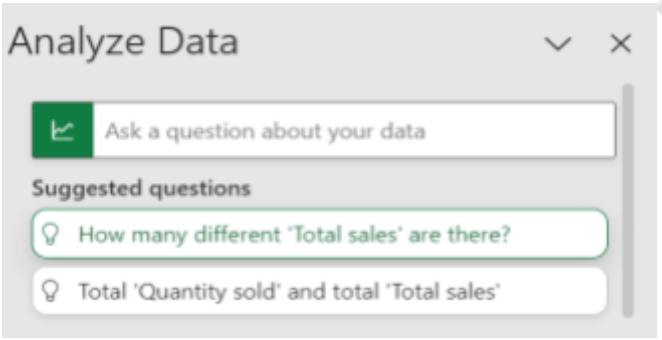


Figure 28*: Representation of the questions it asks and accordingly provides insights*

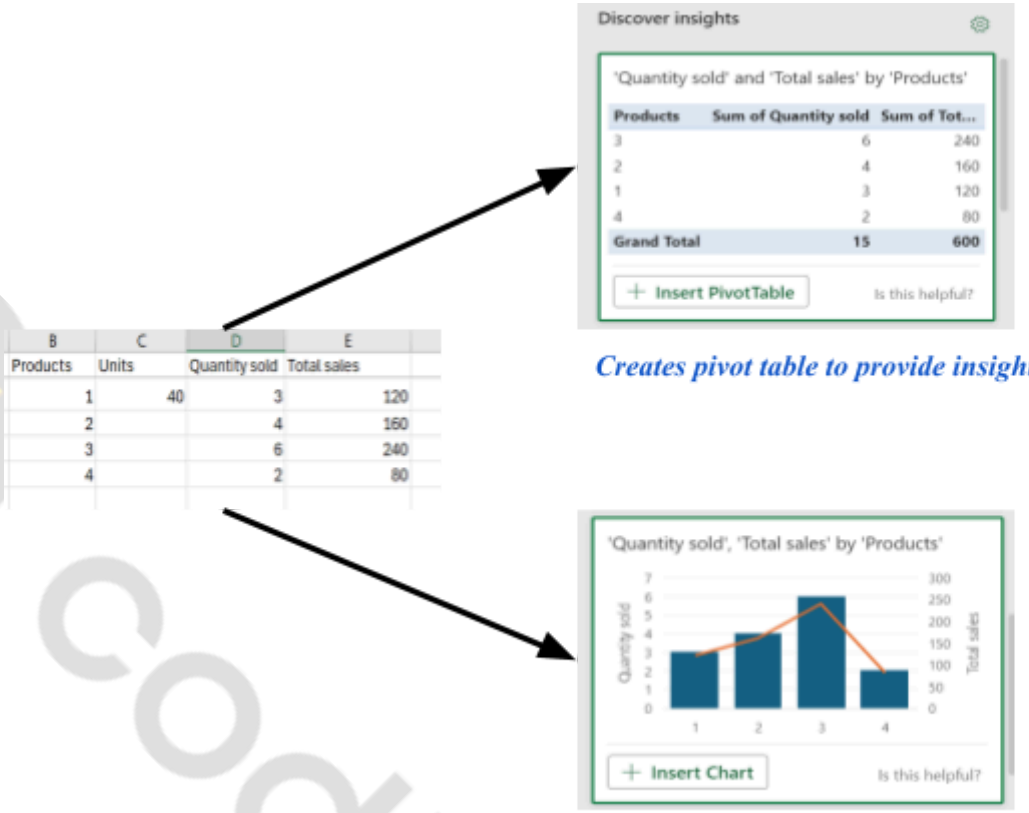


***Figure 29**: Representation of how to analyze data feature provides insights*

2. **Flash fill:**

Flash Fill is a powerful feature in Microsoft Excel that allows users to automatically fill data based on patterns recognized in adjacent cells. It's particularly useful for quickly extracting or formatting data without the need for complex formulas or manual entry.

- **Recognizing Patterns:** Excel's Flash Fill feature analyzes the data entered in a column or row and identifies patterns based on the content of adjacent cells.
- **Automatic Fill:** Once a pattern is recognized, Flash Fill automatically fills in the remaining cells in the column or row with data that follows the same pattern.

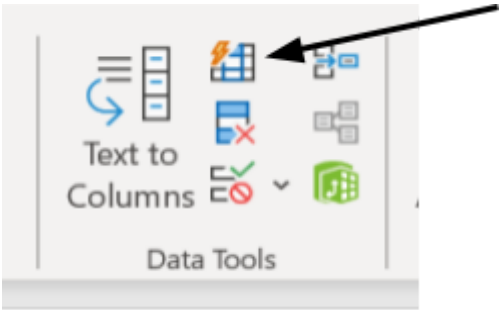


***Figure 30**: Representation of flash fill icon*

3. **Recommended charts:**

The "Recommended Charts" feature in Excel suggests the most suitable chart types based on the selected data.

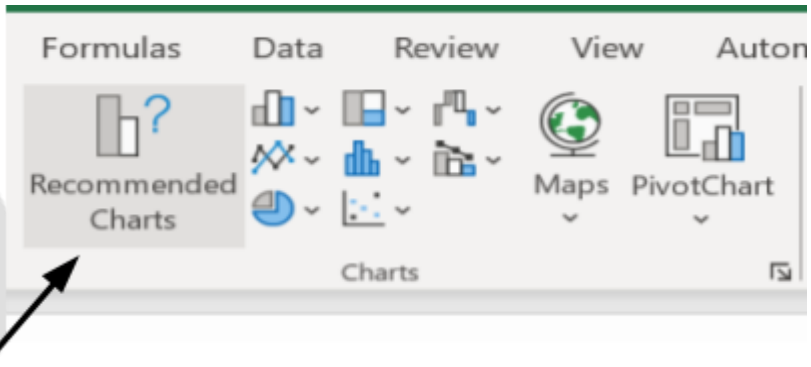


Figure 31: Representation of recommended charts icon

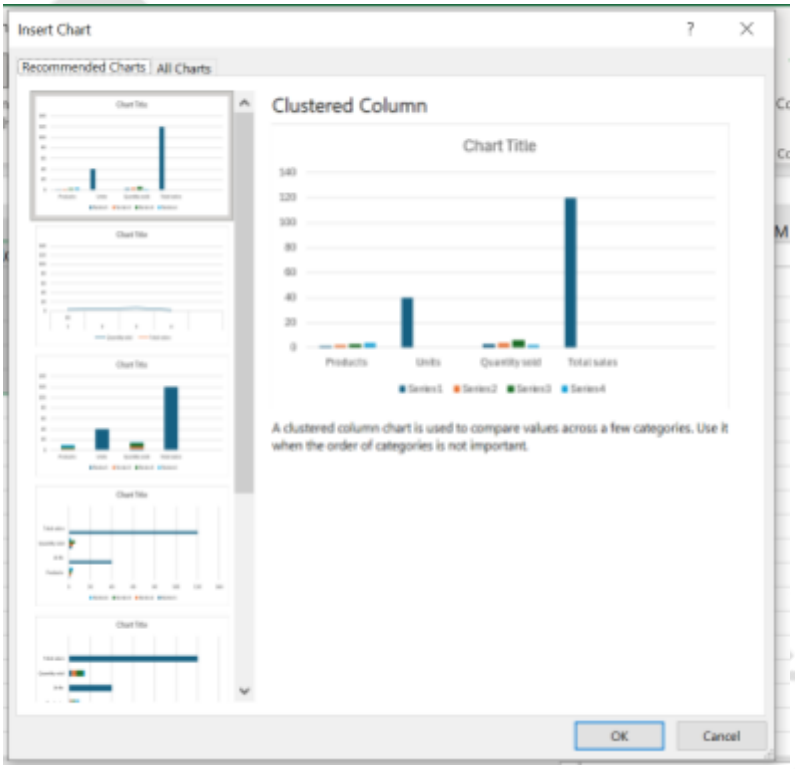


Figure 32: Representation of recommended charts

In-Class Practice Activity:

Dataset - employee_sales.csv

Problem: The Finance Department's "Operational Drag" Audit

The CEO of Elite Dynamics wants to identify "High-Velocity Performers"- employees who are not only generating high revenue but are also ensuring rapid delivery turnarounds. You have been tasked

with auditing the **second half of November** (November 16th to November 30th) to identify transactions that qualify for the "Elite Logistics" badge.

The Challenge:

An entry is only awarded the **"Elite Logistics" badge** if it meets the following three strict conditions:

1. The transaction status is **"Completed."**
2. The **Shipping Gap** (the number of days between the Order Date and Delivery Date) is **4 days or less.**
1. The **Revenue** generated is **greater than $5,000.**

The Task:

2. Identify all transactions that meet the date range (Nov 16 - Nov 30) and the three "Elite" conditions mentioned above.
3. For every employee who qualifies, you must generate a **"Security Passport ID"** in a new column. The ID is created by combining the **first 3 characters of the Employee Name** and the **Year** of the Order Date (e.g., if "Allison Hill" qualified, her ID would be All2024).
4. Calculate the **total count** of employees who qualify for the "Elite Logistics" badge.
5. Identify the **Passport ID** of the individual who achieved the **highest Revenue** among this specific "Elite" group.

Based on your analysis, how many employees qualified for the "Elite Logistics" badge, and what is the Passport ID of the top individual in that group?

OPTIONS:

A) 5 Employees; Ale2024
B) 2 Employees; Jam2024
C) 3 Employees; Bri2024
D) 5 Employees; Car2024

Expected Answer: Option A (5 Employees; Ale2024)